Recall : March 23, 2026

First derivative test

Theorem: Suppose $X \subseteq \mathbb{R}^n$ is open and $f: X \subseteq \mathbb{R}^n \to \mathbb{R}$ has a local extremum at $\vec{a}$ in X, then f has a critical point at $\vec{a}$ (that means either f is not differentiable at $\vec{a}$, or $\vec{\nabla} f|_{\vec{a}} = \vec{0}$).

Second derivative test

Theorem: Suppose $f: X \subseteq \mathbb{R}^n \to \mathbb{R}$ is twice differentiable at $\vec{a}$ in X and $\vec{\nabla} f|_{\vec{a}} = \vec{0}$, then:

1) If $Hf|_{\vec{a}}$ is negative definite, then $f(\vec{a})$ is a local maximum.
   (when $n=2$, $Hf|_{\vec{a}}$ is negative definite if and only if $\det(Hf|_{\vec{a}}) > 0$ and $f_{xx}|_{\vec{a}} < 0$)

2) If $Hf|_{\vec{a}}$ is positive definite, then $f(\vec{a})$ is a local minimum.
   (when $n=2$, $Hf|_{\vec{a}}$ is positive definite if and only if $\det(Hf|_{\vec{a}}) > 0$ and $f_{xx}|_{\vec{a}} > 0$)

3) If $Hf|_{\vec{a}}$ has both a positive and a negative eigenvalue, then f has a saddle point at $\vec{a}$.
   (when $n=2$ $Hf|_{\vec{a}}$ has a positive and a negative eigenvalue if and only if $\det(Hf|_{\vec{a}}) < 0$)

   * For any n: • $Hf|_{\vec{a}}$ has a positive eigenvalue if and only if there is a vector $\vec{v}$ such that $\vec{v}^T Hf|_{\vec{a}} \vec{v} > 0$
   • $Hf|_{\vec{a}}$ has a negative eigenvalue if and only if there is a vector $\vec{v}$ such that $\vec{v}^T Hf|_{\vec{a}} \vec{v} < 0$

4) If none of 1), 2), 3) hold, then the second derivative test is inconclusive.
   (when $n=2$, this happens if $\det(Hf|_{\vec{a}}) = 0$) (In our class, for simplicity when $n > 2$, you can state the test is inconclusive if $\det(Hf|_{\vec{a}}) = 0$)

* The linear algebra facts above work because $Hf|_{\vec{a}}$ is a symmetric matrix

Absolute Extrema

def: A subset $X \subseteq \mathbb{R}^n$ is called $\boxed{\text{compact}}$ if it is
closed (X contains all its boundary points)
and **bounded** (X is contained in some ball of finite radius).

e.g. $X = [-1, 2] \subseteq \mathbb{R}$ is compact
is a closed interval (boundary points are $-1, 2$ which are in X)
and is bounded ($X \subseteq B_3(0) = (-3, 3)$)

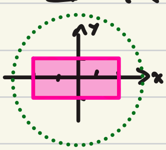
in $\mathbb{R}$
$*$ $B_r(a) = \{x \in \mathbb{R} : |x - a| < r\}$

$X = [3, 5)$ is not compact
is bounded ($X \subseteq B_2(4)$)
but is not closed (5 is a boundary point not in X)

$X = [4, \infty)$ is not compact
is closed (only boundary point is 4 and is in X)
but not bounded

$*$ in $\mathbb{R}^n$ $B_r(\vec{a}) = \{\vec{x} \in \mathbb{R}^n : \|\vec{x} - \vec{a}\| < r\}$

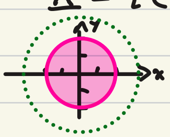
e.g. $X = \{(x,y) \in \mathbb{R}^2 : |x| \leq 2, |y| \leq 1\} \subseteq \mathbb{R}^2$ is compact



is bounded since $X \subseteq B_3(\vec{0})$
is closed since X contains its boundary
the boundary of X is

$$bd(X) = \{(2,y) \in \mathbb{R}^2 : |y| \leq 1\} \cup \{(x,1) \in \mathbb{R}^2 : |x| \leq 2\}$$
$$\cup \{(-2,y) \in \mathbb{R}^2 : |y| \leq 1\} \cup \{(x,-1) : |x| \leq 2\}$$

$X = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 4\} \subseteq \mathbb{R}^2$ is compact



is bounded since $X \subseteq B_3(\vec{0})$
is closed: $bd(X) = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = 4\}$

Theorem (Extreme Value Theorem):
If $f: X \subseteq \mathbb{R}^n \to \mathbb{R}$ is continuous and X is compact, then f attains a global max and a global min on X.
(i.e. there exist $\vec{a}$ and $\vec{b}$ in X such that for all $\vec{x}$ in X

$$\text{global min} \rightarrow f(\vec{a}) \leq f(\vec{x}) \leq f(\vec{b}) \longleftarrow \text{global max} \quad)$$

Moreover, all global extrema must occur either at a boundary point, or at the site of a local extremum (i.e. at a critical point of f).

e.g. Find the global extrema of $f(x) = x^3 - 3x^2$ on $[-2, 2]$

1) Find boundary points of X
2) Find critical points of f inside X
3) Evaluate f on each boundary point and critical point
4) The largest value of f from 3) is the global max, and the smallest value of f from 3) is the global min

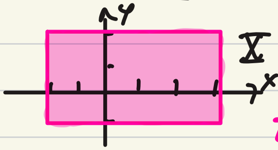
1) $bd(X) = \{-2, 2\}$
2) f is C^1 (continuous derivatives) $f'(x) = 3x^2 - 6x$
$f'(x) = 0 \Rightarrow \quad 0 = 3x(x-2) \quad x=0,\ x=2$

3) $f(-2) = (-2)^3 - 3(-2)^2 = -8 - 12 = -20$
$f(2) = (2)^3 - 3(2)^2 = 8 - 12 = -4$
$f(0) = 0$

4) $f(-2) = -20$ is the global min of f over X

$f(0) = 0$ is the global max of f over X

e.g. Find the global extrema of $f(x,y) = (x-1)^2 + y^2$
over $X = \{(x,y) \in \mathbb{R}^2 : -2 \leq x \leq 3 , -1 \leq y \leq 2\}$



1) $bd(X) = ?$

$\{(3,y) \in \mathbb{R}^2 : -1 \leq y \leq 2\} \cup \{(x,2) \in \mathbb{R}^2 : -2 \leq x \leq 3\}$
$\cup \{(-2,y) \in \mathbb{R}^2 : -1 \leq y \leq 2\} \cup \{(x,-1) \in \mathbb{R}^2 : -2 \leq x \leq 3\}$

2) Critical points: f is continuously differentiable C^1.

$\vec{\nabla} f = (2(x-1), 2y)$

$\vec{\nabla} f = 0 \Rightarrow 2(x-1) = 0, x = 1$ and $2y = 0, y = 0$

critical points: $\{(1,0)\}$ only one critical point

3) Find values of f at boundary points and critical points:

at critical points: at $(1,0)$ $f(1,0) = 0$

at boundary:

over $\{(3,y) \in \mathbb{R}^2 : -1 \leq y \leq 2\}$: $f(3,y) = (3-1)^2 + y^2$
$= 4 + y^2 \quad -1 \leq y \leq 2$

* 4) max and min of $4 + y^2$ over $[-1,2]$:

min at vertex $4 + (0)^2 = 4$

max in one of $4 + (-1)^2 = 5$ or $4 + (2)^2 = 8$

finishing example next class